$$\mathrm{AP@12}(u) = \frac{1}{\min(|Y_u|, 12)} \sum_{k=1}^{12} P_u(k) \times \mathrm{rel}_u(k), \qquad \mathrm{MAP@12} = \frac{1}{|U|} \sum_{u=1}^{|U|} \mathrm{AP@12}(u) \tag{2.7}$$